

You and your roommate are preparing to launch Kristen's Cookie Company's program in your campus apartment. The company will provide fresh cookies to hungry students late at night. You need to evaluate the company's preliminary project in order to make key policy decisions, including the prices to charge, the supplies to order, and the terms of the many orders to accept, and to determine if the business can be profitable.

Business Concept

Your idea is to make fresh cookies to order, using any combination of ingredients that the buyer wants. The cookies will be ready for pickup at your apartment within the hour following the order.

Several factors will distinguish you from competing products such as store-bought cookies. First, your cookies will be completely fresh. You will not bake any cookies until you receive an order, meaning buyers will receive cookies that are literally hot out of the oven. Next, you will have a variety of ingredients available to add to the base dough, including chocolate chips, M&M's, chopped Heath bars, coconut, nuts, and raisins. Buyers will place their orders by phone and specify which of these ingredients they want in their cookies.

You guarantee them cookies that are absolutely fabulous.

The production process

Baking cookies is simple: place all the ingredients in a bowl and mix them; use a cookie scoop on a tray; put the cookies in the oven; bake them; take the tray of cookies out of the oven; let the cookies cool; and finally, remove the cookies from the tray and carefully pack them in a box. You and your roommate already own all the necessary equipment: a professional-quality electric mixer, baking trays, and spoons. Your apartment is equipped with a small oven that can hold one tray at a time. Your landlord pays for all the electricity. The variable costs, therefore, are simply the cost of the ingredients (estimated at \$0.60), the cost of the box in which the cookies are packaged (\$0.10 per box); each box contains a dozen cookies, and your time (whatever value you place on your time).

A detailed examination of the production process, which specifies the duration of each of the steps, is as follows. The first step consists of taking an order, which will be extremely quick and 100% accurate, since your roommate has developed a method using the campus email system to accept orders and inform customers when their orders will be ready for pickup. Because this works automatically on your personal computer, it does not take your time or that of your roommate. Therefore, this step will be ignored in a further analysis.

You and your roommate have timed the necessary physical operations. The first physical operation. The production step involves washing the bowl of the electric mixer and the

beaters from the previous batch, adding the ingredients into the bowl, and turning on the mixer to mix the ingredients. The electric mixer can hold and mix the ingredients to make up to three dozen cookies. Then, you spoon the cookies onto a baking sheet, a dozen at a time. These activities take 6 minutes for the washing and mixing steps, regardless of how many cookies are made in the batch. That is to say, mixing enough dough and ingredients for three dozen cookies takes the same 6 minutes as for a dozen cookies. However, placing the cookies on the tray takes 2 minutes per tray.

The next step, carried out by your roommate, is to put the cookies in the oven and set the thermostat and timer, which takes about 1 minute in total. The cookies bake for the next 9 minutes. Therefore, the total baking time is 10 minutes, during the first minute while your roommate is busy setting the oven. Since the oven can only hold one tray, baking a second dozen trays takes an additional 10 minutes. Your roommate also performs the final steps of the process by first removing the cookies from the oven and setting them aside to cool for 5 minutes, then carefully packing them into a box and accepting payment. Removing the cookies from the oven takes negligible time, but it must be done quickly. It takes 2 minutes to pack each dozen and about 1 minute to accept payment for the order.

This is the process you plan to use to produce cookies by the dozen at Kristen's Cookie Company. As experienced bakers know, some simplifications have been made to the description of the actual cookie production process. For example, the first batch of cookies for the night requires preheating the oven. However, these complexities will be set aside for now.

- 1- Start your analysis by creating a process diagram of the cookie manufacturing process.

- 2- Key questions:

To launch the business, you need to set prices and establish rules for accepting orders. Problems will only be resolved once you have started and tried different ways to produce the cookies. Before you begin, however, you should at least establish a preliminary plan, specifying as much as possible what you want to do, so that you can accurately calculate the time you will need to dedicate to this activity each evening, and how much money you can expect to earn. For example, when you organize a market to determine the likely demand, you will want to specify exactly what your ordering policies will be. Therefore, the answers to the following operational questions should help you:

1. How long will it take you to fill an urgent order of 12 cookies? Of 24 cookies of the same variety?
2. How many orders can you fill in one night, assuming you are open for four hours each night?

3. How much of your precious time and your roommate's will it take to fill each order?
4. Since your baking trays can hold exactly a dozen cookies, you will produce and sell cookies by the dozen. Do you offer a discount to people who order two dozen of the same cookies, three dozen of the same cookies, or more? If so, how much?
5. How many electric mixers and baking trays will you need?
6. Are there any changes you can make to your production plans that will allow you to make better cookies or more cookies in less time or at a lower cost? For example, is there a bottleneck in your production process that you can cheaply develop? What is the effect of adding another oven?
7. What happens if you try to do the work alone?
8. When do you need to promise delivery? How can you quickly check your order sheet (list of current orders) and tell a caller when their order will be ready? How much of a safety margin for the schedule should be anticipated?
9. What other factors do you need to consider at this stage of planning your business?
10. Your product must be made to order as each order is potentially unique. If you decide to sell standard cookies instead, how would you change the production system? The order-taking process? Other policies?